

Úkol 3.2

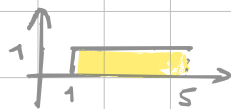
a)

$$(v_1, v_2, v_3) = (1, x, x^2)$$

$$\|v_1\| = \sqrt{\int_1^5 1 \cdot 1 \, dx} = \sqrt{4}$$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2}$$

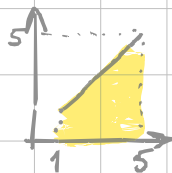
$$u_1 = \frac{1}{2}$$



2) chci w_2 OĚ projekci v_2 na $\text{span} \{u_1\}$

$$w_2 = \langle u_1, v_2 \rangle \cdot u_1 = \int_1^5 \frac{1}{2} \cdot x \, dx \cdot \frac{1}{2}$$

$$= \frac{1}{2} \cdot 12 \cdot \frac{1}{2} = 3$$



kolmice $v_2 - w_2 = x - 3$

$$\|v_2 - w_2\| = \sqrt{\int_1^5 (x-3)^2 \, dx} = \sqrt{\int_1^5 x^2 - 6x + 9 \, dx}$$

$$= \sqrt{\frac{5^3}{3} - \frac{1}{3} - 3 \cdot 5^2 + 3 \cdot 1 + 9 \cdot 5 - 9 \cdot 1} = \sqrt{\frac{16}{3}}$$

$$u_2 = \frac{x-3}{\frac{4}{\sqrt{3}}} = \frac{\sqrt{3}}{4} (x-3)$$

$$u_2 = \frac{\sqrt{3}}{4} (x-3)$$

3) w_3 je OĚ projekce v_3 na $\text{span} \{u_1, u_2\}$

$$w_3 = \langle u_1, v_3 \rangle \cdot u_1 + \langle u_2, v_3 \rangle \cdot u_2$$

$$= \int_1^5 \frac{1}{2} \cdot x^2 \, dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 \, dx \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$= \frac{1}{2} \cdot \frac{124}{3} \cdot \frac{1}{2} + \frac{\sqrt{3}}{4} \cdot 32 \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$= \frac{31}{3} + 6x - 18 = 6x - \frac{23}{3}$$

kolmice je $v_3 - w_3 = x^2 - 6x + \frac{23}{3}$

$$\|v_3 - w_3\| = \sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 dx} = \sqrt{\frac{256}{45}}$$

$$u_3 = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16}{3\sqrt{5}}} = \frac{3\sqrt{5}x^2 - 18\sqrt{5}x + 23\sqrt{5}}{16}$$

b) Og projekce je vektor $(c, b, a)^T$

přičemž $c = \langle u_1, \sin x \rangle$, $b = \langle u_2, \sin x \rangle$, $a = \langle u_3, \sin x \rangle$

$$c = \int_1^5 \frac{1}{2} \cdot \sin x \, dx \approx 0,128$$

$$b = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x \, dx \approx -1,493$$

$$a = \int_1^5 \frac{3\sqrt{5}x^2 - 18\sqrt{5}x + 23\sqrt{5}}{16} \cdot \sin x \, dx \approx -0,125$$

Og projekce $\sin x$ na $\text{Span} \{1, x, x^2\}$ je
 $0,128 - 1,493x - 0,125x^2$

$$\underline{\underline{c)}} \quad \|v\| = \sqrt{\langle v, v \rangle}$$

$$\| \sin x - (ax^2 + bx + c) \|^2 = \int_1^5 (\sin x - (ax^2 + bx + c))^2 dx$$

$$= \int_1^5 (\sin x + 0,125x^2 + 1,493x - 0,128)^2 dx$$

$$\approx 143,996$$

$$\text{proto} \quad \| \sin x - (ax^2 + bx + c) \| = \sqrt{143,996} \approx 11,9998$$

$$\approx 12$$